

A Surface-Content Decomposition of AI-Generated Text Detection Benchmarks

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Abstract—A detector scoring 99% on a machine-generated-text benchmark may have learned how machine language differs from human language. It may instead have learned how one corpus was punctuated. The headline number does not separate the two. We decompose the signal into three arms. A surface-only model reads 47 orthographic features and never sees a word. A content-only model reads text stripped of punctuation, casing and non-ASCII characters. A fine-tuned transformer on raw text is the reference. On HC3 the two arms are indistinguishable under a paired test, 3.20% against 3.26% error, and the null holds on five of five partitions with the sign changing between them. DeBERTa reaches 0.28% on the same split, so the parity joins two weak arms rather than a ceiling. On DAIGT V2 content is stronger by a factor of 7.9 in error. Subgroup splits localise that result rather than confirm it. The surface arm reaches 0.01% error on the Reddit sub-corpus, which is 74.8% of HC3, and loses to content on both other domains large enough to test. Removing the cue responsible, the rate of spaces before punctuation, costs that arm 10.03 points. BERT emits identical token identifiers for strings differing only by that cue, so it cannot represent it, and it still reaches 0.9916 weighted F1. Surface form is informative on both corpora. We claim only that it reaches parity with content on one, and there for one reason.

Index Terms—AI-generated text detection, benchmark evaluation, surface features, shortcut learning, tokenisation

I. INTRODUCTION

Detectors of machine-generated text report accuracy at or above 97% on the benchmarks the field uses most, and often above 99% [1], [2], [3]. Read directly, those numbers say the task is close to solved. Read carefully, they say something weaker. A benchmark score measures the separability of a particular corpus, not the capability a reader infers from it.

Separability has more than one source. A detector can reach a high score by modelling how machine-generated language differs from human language. It can also reach it by modelling how one corpus was assembled, punctuated and encoded. Both produce the same headline figure. Individual instances of the second are documented. They include a whitespace convention in HC3 [4], a length confound in the same corpus [5] and prompt-specific collection shortcuts [6]. What is missing is a way to ask how much, in total, a benchmark’s separability owes to surface form rather than content, and to place two benchmarks on the same axis.

Three lines of prior work come close, and each stops short. Shared benchmarks such as RAID [7] and M4 [8] make detectors comparable across conditions, but do not ask what any one condition is separable by. Cleaning kits remove a single known cue [4], but a null after cleaning says nothing unless the detector could read the cue. Partial-input baselines

expose artefacts in natural language inference [9], [10], but have not been transferred to detection benchmarks, where the artefact is a property of the text rather than a span of it.

This paper proposes that measurement. We compare three arms on identical splits. A *surface-only* model reads 47 orthographic features and never sees a word. A *content-only* model reads text after punctuation, casing and non-ASCII characters are stripped. A *full* model is a fine-tuned transformer on raw text. The first two share a classifier family and are directly comparable. The third is a reference point, not a matched arm. The object of measurement is a corpus, not a detector, and we offer no detector comparison, since a fair one would need matched training data on both corpora. On HC3 the two arms are indistinguishable, 3.20% against 3.26% test error, while on DAIGT V2 content is stronger by a factor of 7.9. Splitting each corpus by its own subgroup labels then shows the HC3 tie belongs to the Reddit sub-corpus that is three quarters of it, and reverses on the other domains. The main contributions are the following.

- A surface-content decomposition, stated formally in Section III-D as a reusable measurement over any human-versus-machine corpus, with the length control that stops its two arms sharing a channel.
- Its application to two benchmarks and their eighteen sub-corpora (Sections IV-B and IV-C). This locates the HC3 result in one collection convention in one dominant domain, and shows that removing that convention costs the surface arm ten points.
- A tokenisation control (Section IV-D) establishing that the most-discussed cue in HC3 is sufficient in isolation yet unnecessary in practice, since BERT cannot represent it and still reaches 0.9916 weighted F1 there.

The rest of this paper is organised as follows. Section II reviews detection on these benchmarks, shortcut learning and robustness. Section III states the problem setting, the corpora, the decomposition and the controls. Section IV reports the results. Section V discusses what they license, states the limits and concludes.

II. RELATED WORK

Detection on these benchmarks. HC3 was introduced together with a RoBERTa detector at 99.82 F1 [1]. That established both the corpus and the accuracy regime the field has worked in since. The regime narrows to roughly 91.7% under semantic-invariant tasks [2]. Competitive DAIGT V2 accuracy has been reached with character n-grams and

no transformer [3]. That result is consistent with ours, but it was reported without the surface-content separation that would explain it. DeBERTa has been paired with stylometric features [11], and style embeddings alone have been used to detect [12]. Both combine the two channels to raise a score, where we hold them apart to measure one.

The closest prior result. The HC3 whitespace convention was identified in [4], which released a cleaning kit. Its appendix reports that a detector consisting of a single test for one token identifier reaches 82.12 F1 on HC3 at sentence level, above the 81.89 it quotes for a fine-tuned RoBERTa. Our document-level equivalent of that rule reaches 94.22 weighted F1. The difference is sentence against document, not a disagreement. That work is the closest existing statement of this paper’s HC3 finding.

Shared benchmarks and shortcut learning. RAID [7], M4 [8] and SemEval-2024 Task 8 [13] make detectors comparable across generators, domains and attacks. They do not ask what any one condition is separable by. That is the axis added here, and the reason this measurement would apply to them unchanged. The pattern is not specific to text detection. Vision datasets are identifiable from their own images [14]. In natural language inference, a model reading only the hypothesis reaches far above chance [9], [10], and [15] gives the general account. Our surface arm is a partial-input baseline transferred to this task, except that it removes a property of the text rather than a span of it. The caveat that governs how such a baseline may be read comes from [16]. A partial-input model scoring highly shows a dataset is cheatable. One scoring poorly does not show it is clean, since the restricted representation may fail to express an artefact that is present. We therefore report the surface arm as an upper bound on surface-carried separability, not as a measurement of it.

Robustness and fairness. HC3-trained detectors attacked with misspellings and homoglyphs fall from 99.88 F1 to 33.57% accuracy [17]. Deployed detectors misclassify non-native English writing as machine-generated at high rates [18]. That bears on our result, since a detector whose separability rests on orthographic regularity should concentrate its errors on writers whose habits differ from the corpus it was fitted to.

The gap. None of the above measures how much of a benchmark’s separability is carried by surface form. Unlike [4], which removes one cue, and [7], [8], which compare detectors across conditions, this paper measures each corpus with two matched arms, locates the result per sub-corpus, and checks whether the detector can read the cue at all.

III. METHOD

A. Problem setting

The object of measurement is a corpus. Given a human-versus-machine benchmark, we ask how much of its separability a model could obtain without reading the language. The instruments are two matched classifiers over disjoint views of each document, plus a fine-tuned transformer as a reference (Fig. 1). We assume balanced classes, English text and a 128-

token transformer budget, and make no claim about which detector is best.

B. Corpora and partitioning

DAIGT V2 [19] contains 44,868 argumentative student essays, 27,371 human-written and 17,497 machine-generated by a mixture of 2023-era systems, assembled for a public detection competition. HC3 [1] contains 85,449 question-answer rows from five English sources, contrasting human answers with GPT-3.5-Turbo. The two differ in nearly every respect that matters here. DAIGT V2 has many generators, one genre and long documents. HC3 has one generator, five source domains and short ones. Both were class-balanced by downsampling, giving 34,994 and 53,806 rows. That fixes a degenerate single-class predictor at 0.333 weighted F1 rather than something respectable.

HC3 carries 6,118 duplicate rows, 7.16% of the corpus, so the split is group-aware. Documents are grouped by an MD5 hash of their whitespace-normalised, lowercased text, and whole groups go to one partition. The partition is 72/8/20, since a fifth goes to test first and a tenth of the remainder to validation. Measured directly, the group-aware rule leaks 0 of 10,732 HC3 test documents, where a plain stratified split of the same sample leaks 570, or 5.30%. The split is built once and reused by every model and seed.

C. Models

Five families are evaluated. Three are classical, Naive Bayes, logistic regression and a linear support vector machine, each under bag-of-words and TF-IDF. Two are transformers, `bert-base-uncased` [20] and `microsoft/deberta-v3-base` [21], fine-tuned over a sixteen-run grid per dataset varying learning rate, batch size and weight decay, with the operating point selected on validation weighted F1. DeBERTa’s SentencePiece tokeniser [22] encodes leading whitespace and BERT’s WordPiece does not. That makes the pair a controlled contrast on exactly the cue Section IV-D examines.

D. The decomposition

Write x for a document and $y \in \{0, 1\}$ for its label, with 1 denoting machine-generated. The surface map ϕ_S sends a document to $\mathbb{R}^{47}$, a vector of orthographic statistics covering punctuation rates, whitespace behaviour including spaces before punctuation, casing, document and sentence length, non-ASCII and digit rates. It reads no word identity. The content map ϕ_C lowercases, replaces every character outside the lowercase Latin alphabet and whitespace with a space, collapses runs of whitespace, then takes raw bag-of-words counts. Punctuation, casing and non-ASCII cannot enter it.

Each arm fits one logistic regression on the same training partition,

$$h_a = \arg \min_h \sum_{(x,y) \in \mathcal{D}_t} \ell(h(\phi_a(x)), y) + \frac{1}{2C} \|h\|_2^2, \quad (1)$$

for $a \in \{S, C\}$, with C fixed at the library default for both. Sharing the classifier family and regularisation is what makes

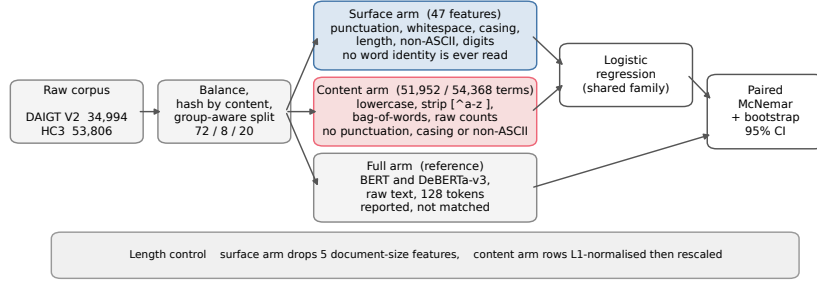


Fig. 1. The measurement pipeline. The surface and content arms read disjoint views of every document and share one classifier family. The transformer is a reference, not a matched arm.

the two arms comparable. Section IV-B reports what changes when C is chosen on validation for each arm instead. We report the held-out error ε_a of each arm and the gap $\Delta = \varepsilon_S - \varepsilon_C$. A gap near zero says orthography alone reaches the accuracy of content alone. The comparison is between two specific arms, not a partition of the total separability, so a richer featurisation would lower either side, and the gap is not a claim that a corpus is clean.

One channel reaches both arms. The surface arm reads size counts directly, and raw bag-of-words rows sum to document length. Length is a documented confound on both corpora [5], so we close it. We drop the five size-scaling surface features and normalise content rows to unit total before rescaling by the mean training document length. The rescaling matters. Rows summing to one shrink every feature, and the arm would otherwise collapse for reasons of scale rather than length.

E. Statistics and controls

Transformer baselines are read from the deployed checkpoint’s own record, never from a grid maximum. Where two models are compared on the same test set the comparison is paired. Writing b and c for the discordant counts, we report the exact two-sided binomial p -value [23] together with a paired bootstrap 95% interval on the error difference over 10,000 resamples [24], resampling document indices so the pairing is preserved. A bootstrap says nothing about how the corpus was partitioned, and the central claim here is a null. The decomposition is therefore repeated over five group-aware partitions varying only the split seed. We run no parametric tests at $n \leq 30$.

Two controls guard the claims below. A *tokenisation* control compares token identifier sequences for minimally different strings. Whether a detector can exploit a character-level cue is a property of its tokeniser, and it is directly testable. A *pipeline-fires* control confirms that a cleaning step reporting no change actually executed. It applies the same pipeline to the corpus where the step removes something the tokeniser can represent.

F. Reproducibility

Partitions are built once at split seed 42 and reused by every model and every training seed, so only initialisation and batch order vary between runs. Re-splitting per seed would change the evaluation set and invalidate the comparison. The four

TABLE I
TEST ERROR FOR ALL EIGHT CONFIGURATIONS. TRANSFORMER ROWS ARE DEPLOYED CHECKPOINTS, NEVER GRID MAXIMA. BOLD MARKS THE BEST ENTRY PER CORPUS AND UNDERLINE THE BEST CLASSICAL ONE.

Model	Rep.	DAIGT V2 err %	HC3 err %
Naive Bayes	BoW	4.09	12.87
Naive Bayes	TF-IDF	4.23	13.28
Logistic Reg.	BoW	1.07	4.49
Logistic Reg.	TF-IDF	1.43	6.35
SVM	BoW	1.30	5.25
SVM	TF-IDF	<u>0.90</u>	5.51
BERT	raw	0.84	0.84
DeBERTa	raw	0.83	0.28

deployed checkpoints were retrained at seeds 123 and 456 for the ranges in Section IV. Fine-tuning ran three epochs at a 128-token budget on one 8 GiB consumer GPU, the longest run taking 1,111 seconds at 4.22 GiB peak memory, under PyTorch 2.11, Transformers 4.57 and scikit-learn 1.9. Code, split manifests and the per-run record behind every number are kept in a public repository, withheld here for anonymity.

IV. RESULTS

A. A classical model matches a transformer on DAIGT V2 until the text budget is matched

Table I reports every configuration, and the two benchmarks part company. On DAIGT V2 the best classical configuration sits 0.07 error points above the best transformer. On the same documents the two disagree on 99 cases split 47 to 52, so McNemar returns $p = 0.69$, and the paired interval on the error difference, $[-0.21, +0.34]$ points, contains zero. On HC3 the same comparison gives 484 discordant cases split 16 to 468, $p < 10^{-6}$, and an error difference of +4.21 points.

Reseeding the four deployed checkpoints at 123 and 456 moves them very little. DAIGT V2 BERT averages 0.9921 over a range of 0.0011 and DeBERTa 0.9930 over 0.0024, so the two overlap. HC3 BERT averages 0.9930 over 0.0036 and DeBERTa 0.9970 over 0.0005, and those ranges are disjoint. The tie and the separation both survive reseeding.

That tie is also not a matched comparison. The classical models read whole documents, while the transformers read 128 tokens, which is 33.5% of a mean DAIGT V2 essay against 74.6% of a mean HC3 answer. We therefore refit the classical grid on the exact character span each checkpoint’s tokeniser kept, taken from offset mapping. The best DAIGT V2 configuration moves from 0.90% to 2.10% error against

TABLE II

TOP, THE DECOMPOSITION ON EACH CORPUS, WITH THE LOWER ROWS CLOSING THE DOCUMENT-LENGTH CHANNEL BOTH ARMS SHARE. BOTTOM, HC3 BY SOURCE DOMAIN, EACH REBALANCED TO PARITY, WITH THE SHARE OF DOCUMENTS CARRYING A SPACE BEFORE PUNCTUATION. BOLD MARKS THE LOWER ERROR OF THE TWO ARMS.

Arm	DAIGT V2 err %		HC3 err %	
surface-only	7.86		3.20	
content-only	0.99		3.26	
full transformer	0.83		0.28	
length-only (3 feats)	24.59		15.23	
surface-only, no length	9.93		3.37	
content-only, length-norm.	0.94		6.28	

HC3 domain	<i>n</i>	surface		content		cue present in	
		<i>n</i>	err %	<i>n</i>	err %	human	machine
reddit_eli5	6,690	0.01	2.66	98.9%	0.3%		
finance	684	7.60	3.07	5.6%	0.1%		
medicine	250	4.00	0.40	17.0%	0.1%		
open_qa	198	4.04	5.05	60.7%	0.4%		
wiki_csai	168	5.36	11.90	1.6%	0.2%		

open_qa and wiki_csai fall below 200 test rows.

BERT’s 0.84%, a difference of 1.26 points with interval $[-0.90, +1.63]$, and the DeBERTa window returns the same difference. On HC3 the same refit widens the gap from 4.49% to 5.66%. Held to one text budget the transformers lead on both corpora. The unmatched tie was a fact about information access, not about architecture.

B. On HC3, orthography alone matches content alone

Table II gives the decomposition. On HC3 the two arms are statistically indistinguishable, 3.20% error from orthography against 3.26% from content. They disagree on 625 documents split 316 to 309, so McNemar returns $p = 0.81$ and the difference is -0.07 points with interval $[-0.52, +0.40]$. Forty-seven features that never read a word do as well as a bag-of-words model over the whole vocabulary. On DAIGT V2 the same arms separate by a factor of 7.9, 7.86% against 0.99%, with 569 discordant documents split 44 to 525 at $p < 10^{-6}$.

The content arm is deliberately unfiltered. Refitting it with the stopword removal and lemmatisation of Table I’s classical pipeline reproduces that pipeline exactly, and on HC3 the filtered arm then loses to orthography by 1.30 points where the unfiltered arm ties it. We report the unfiltered arm because it is the stronger opponent, which makes the parity claim harder. Against it the HC3 classical-to-transformer error ratio is 11.6 to one, not the sixteen to one Table I implies.

Surface form is informative on both benchmarks, since DAIGT V2’s surface arm reaches 0.9214 weighted F1. The finding is not that one corpus is clean. It is that only on HC3 does surface rise to parity with content. Closing the length channel sharpens both conclusions. It widens DAIGT V2’s content advantage from 7.9 to 10.6 times and turns HC3’s parity into a 2.91-point advantage for orthography. A null on one partition is what a lucky split manufactures, so the decomposition runs again over five group-aware partitions. Content leads on DAIGT V2 on all five at $p < 10^{-6}$. On HC3 the difference reaches significance on none, with

p of 0.81, 0.27, 0.34, 0.23 and 0.78, and it changes sign between partitions. That is the strong form of the null, since an underpowered real difference keeps its sign.

Both arms take the library default inverse regularisation strength, which makes them comparable to each other rather than to their own best. Choosing that value on validation for each arm separately selects 100 for surface and 0.1 for content on both corpora and changes no conclusion. On HC3 the arms move to 3.15% and 3.47% error and remain indistinguishable on all five partitions, at p from 0.19 to 0.75 with every interval containing zero. The surface arm is then nominally ahead on all five by 0.08 to 0.32 points, so the sign argument above belongs to the default fit and the tuned fit rests on the intervals alone. On DAIGT V2 content keeps its advantage on all five at $p < 10^{-6}$, 0.96% against 7.66% on the main partition.

C. The parity belongs to one sub-domain and one cue

Both corpora carry subgroup labels, and rejoining them to the existing splits recovers a source for every row. The lower half of Table II runs the same measurement per HC3 domain. On reddit_eli5 the surface arm reaches 0.01% error, roughly one mistake in 6,690 documents. On the two other domains with enough test rows to support a claim the ordering reverses and content wins. It wins by 4.53 points on finance with interval $[-2.34, +6.87]$ and by 3.60 on medicine with interval $[-1.60, +6.00]$, neither containing zero. Since reddit_eli5 is 74.8% of the balanced corpus before the per-domain rebalancing this table applies, the corpus-level parity is substantially that one domain.

The mechanism is measured rather than inferred. The space-before-punctuation cue appears in 98.9% of reddit_eli5 human documents against 0.3% of machine ones, and decays across the rest to 1.6% on wiki_csai. A single boolean rule, treating a document as human when the cue is present, scores 99.22% accuracy on reddit_eli5 and 94.23% on the balanced corpus at document level, 94.22 weighted F1. The sentence-level equivalent reported in [4] reaches 82.12 F1. Applying that work’s cleaning kit, and nothing else, changes 44.5% of HC3 documents. It moves the surface arm from 3.20% to 13.22% error, a loss of 10.03 points with interval $[-10.69, -9.38]$, while the content arm is untouched at 3.26%. After cleaning, orthography sits 9.96 points behind content on the corpus where it had tied.

The same treatment on DAIGT V2 pairs each machine generator against the shared human pool. Content wins on every generator. The surface arm’s error, however, spans sixteenfold across the ten generators with enough test rows, from 0.50% to 8.13%. Two pairs of the same underlying model contributed by different people differ by factors of 4.1 and 2.3. We report those as suggestive rather than established, since the smaller member of each pair holds 184 and 130 test rows, below the 200-row floor used elsewhere. On this evidence, surface separability appears to track collection provenance more closely than model identity, which is what the HC3 domain split finds by another route.

D. A model that cannot represent the cue reaches 0.9916 anyway

The natural inference from the cue’s dominance is that HC3-trained detectors exploit it. Tokenisation makes that testable, and the test does not support it. BERT’s WordPiece tokeniser splits on punctuation irrespective of adjacent whitespace. It emits identical identifier sequences for "the answer is simple ." and "the answer is simple." in three of three pairs tested, while DeBERTa’s SentencePiece distinguishes all three. BERT cannot represent the cue and still reaches 0.9916 weighted F1 on HC3. The cue is therefore sufficient in isolation, since the surface arm that reads it reaches 0.9680, and unnecessary in practice.

This is also why removing the cue changes nothing for BERT. Whitespace cleaning on HC3 yields predictions bit-identical to the uncleaned run, which alone would be indistinguishable from a step that never executed. The same pipeline on DAIGT V2, where it also removes emoji that WordPiece represents, does change BERT’s predictions, so the pipeline fires and the null is specific to whitespace. We do not attribute the BERT-to-DeBERTa difference on HC3 to this cue, since the two models differ in tokeniser, architecture, pretraining corpus and parameter count. What the experiment establishes is that a null from removing a cue is uninterpretable without first checking the model could read it. Fig. 2 collects the results above.

E. Neither corpus transfers to the other

Each deployed checkpoint was also evaluated on the corpus it was not trained on, without adaptation, over three seeds. Every cell loses between 0.0833 and 0.2019 mean weighted F1. Trained on DAIGT V2, BERT falls from 0.9921 to 0.7902 (range 0.770 to 0.810) and DeBERTa from 0.9930 to 0.9096 (0.891 to 0.926). Trained on HC3, BERT falls from 0.9930 to 0.8311 (0.813 to 0.852) and DeBERTa from 0.9970 to 0.8512 (0.830 to 0.887). DeBERTa’s mean gap is roughly half BERT’s in both directions. We report that as a mean-level pattern, not a confirmed effect, since BERT’s two ranges nearly touch and three runs do not support a test. Much of what each model learned is specific to the corpus it was fitted to, which is what the decomposition anticipates when separability is carried by how a corpus was assembled.

F. A published detector out of domain

Every model above is one we trained. The RoBERTa detector released with HC3 [25] gives an external reference point, run over the same test partitions without fine-tuning. On HC3 it reaches 0.9952 weighted F1, which is contaminated, since our test rows were almost certainly in its training data. On DAIGT V2, which it has never seen, it reaches 0.8230 weighted F1 at 17.55% error, against 0.83% for our in-domain DeBERTa. One model was trained on the corpus being scored and the other was not, so the gap is not an architecture result. The published detector reads 512 tokens against our 128, which if anything favours it, and we ran it at its own decision boundary. The figure bounds how far the 99.82 F1 quoted in Section II

travels off its own corpus untuned, not how far it could travel in principle.

V. DISCUSSION AND CONCLUSION

The decomposition separates two benchmarks that a headline accuracy figure does not, and applied per sub-corpus it separates one benchmark from itself. On HC3 orthography alone matches content alone on all five partitions. On DAIGT V2 content is stronger by a factor of 7.9. Section IV-C narrows the first of those to one collection convention in the Reddit sub-corpus that is three quarters of the corpus, and removing that convention costs the surface arm ten points. A single collection convention can be cleaned. The diffuse stylistic signal DAIGT V2’s surface arm reads has no equivalent single cleaning step in our experiments, and it may not be an artefact at all.

What the measurement does not license is a claim that either corpus is clean. DAIGT V2’s surface arm reaches 0.9214 weighted F1, so surface form is substantially informative on both. The reason a low surface score would not have settled the question either is given in [16]. The finding is comparative and bounded by the arms we built.

Two results here say more about method than about these corpora. A cleaning experiment reporting no change is uninterpretable without evidence the pipeline ran, and a tokeniser that cannot represent the cue being cleaned makes the null a fact about the model rather than the corpus. Both checks cost one forward pass, and neither is routinely reported.

Several limits bound every number above. The transformers see at most 128 tokens, so the DAIGT V2 transformer results describe classification from an essay’s opening, which is why the classical comparison is repeated on that span in Section IV. The four deployed checkpoints were run at three seeds and move by 0.0005 to 0.0036 weighted F1, but the grid behind them is single-seed. The content arm is bag-of-words, so word order and syntax lie outside it. The surface set is 47 hand-built features, so ε_S is an upper bound. Five of the eighteen sub-corpora fall below 200 test rows after rebalancing and are reported as underpowered rather than as nulls. HC3 has one generator collected at one time, and both corpora are English, so the orthographic conventions the surface arm reads are not language-independent.

The measurement itself is the portable part. It needs no new annotation and fits two logistic regressions. It yields one number per corpus, and per sub-corpus, that a headline accuracy cannot express, namely how much of that corpus a model could pass without reading the language. We would encourage reporting it alongside any new detection benchmark, as a hypothesis-only baseline is now reported alongside a natural language inference dataset [10]. Three extensions follow directly. The first is to run the decomposition per condition on RAID, M4 and SemEval-2024 Task 8, which already carry the labels it needs. The second is to repeat the transformer comparison at a 512-token budget, so the DAIGT V2 results describe whole essays. The third is to widen both arms, adding word order to the content arm and learned

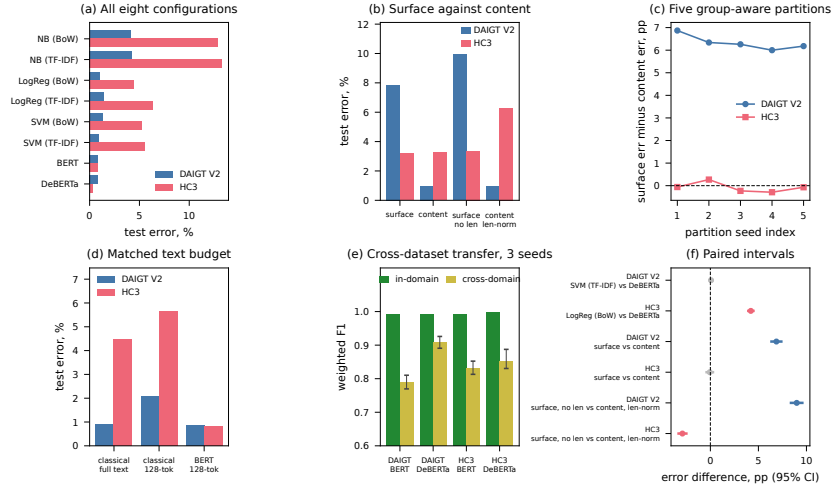


Fig. 2. Every headline result in one view. Panels give the model ranking (a), the decomposition with and without the length channel (b), the surface-minus-content difference across five partitions (c), the matched text budget (d), cross-corpus transfer (e), and the paired intervals behind each significance claim (f), where grey marks an interval containing zero.

character features to the surface arm, and to test a non-English corpus, where the conventions measured here need not hold.

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